Bound Preservation with Sign-Changing Sources, Consistent Dirichlet Boundary Fluxes, and Cross-Code Validation in Thermalio

Thermalio research notes — follow-up to Directions A, B and C

Abstract

We close three open items of the Thermalio research backlog. **(B)** The global maximum-principle limiter of the algebraic flux correction (AFC) scheme used bounds detected from initial and boundary data, which are invalid once a source of either sign moves the solution outside that range. We add a *source envelope* limiter whose time-dependent bounds are proved by a discrete comparison principle, and a *source-local* limiter whose bounds come from the source-including low-order solution. The envelope limiter is exactly as accurate as the unlimited high-order scheme (relative error 2.38×10^{-4} vs. 4.60×10^{-3} for the previous limiter) and never leaves the rigorous bounds, which the high-order scheme violates by 0.063. **(C)** We reconstruct the heat flowing through Dirichlet boundaries as the residual of the constrained boundary-cell equations. This makes the energy audits of the cryosurgery and quench scenarios *non-tautological*: the previous audits inferred the boundary energy from the enthalpy change and were zero by construction; the new audits show closure errors of 26% and 32% on a 24-cell mesh, which we decompose exactly into a latent-heat jump in the Dirichlet band ($L\times$ band area), an interior apparent-capacity defect that vanishes with Δt, and unconverged Picard residuals. The enthalpy solver closes to 10^{-16}. **(A)** We validate Thermalio against two independent external codes, FiPy 4.0.3 (finite volumes) and scikit-fem 12.0.2 (P1/P2 finite elements), on isotropic, anisotropic and Robin/Pennes manufactured problems, and the enthalpy solver against Neumann's classical Stefan solution. All codes agree and converge at their nominal rates on isotropic problems. On the anisotropic problem, Thermalio's reconstructed flux stays near second order while FiPy's tensor diffusion drops to order 0.6–0.9. The study also exposed a limitation of Thermalio itself: on skewed meshes the anisotropic case fell to about first order. **(B′)** We trace that loss to the `reconstructed` flux (averaged linearly exact gradients, no compact coupling). We replace it with a new *quadratic* flux: a 2-exact least-squares reconstruction from cell averages, face-midpoint gradients and a consistent compact jump penalty. The new flux reproduces quadratic solutions to 10^{-14} on every mesh tested and restores second order on skewed (2.00), hexagonal (1.93), randomly perturbed (2.03) and tiled mixed-polygon (1.92) meshes. On the last three, every previous option stalls at order 0.2–1.1.

Contents

1 Bound preservation with sign-changing sources (Direction B, PR5) 2
- 1.1 Setting and the AFC scheme . . . 2
- 1.2 The source envelope . . . 2
- 1.3 Source-local bounds . . . 3
- 1.4 Results . . . 3

2 Consistent Dirichlet boundary fluxes (Direction C, capability 7) 4
- 2.1 Reaction flux . . . 4
- 2.2 Audit identity for the apparent-capacity Cattaneo–Stefan scheme . . . 5
- 2.3 Results . . . 5

3 Cross-code validation against external codes (Direction A) 6
3.1 Codes and protocol 6
3.2 Findings 7
3.3 Neumann's Stefan problem 8

4 Restoring second order: the quadratic flux 8
4.1 Diagnosis 8
4.2 Formulation 8
4.3 Results 9

5 Limitations and next steps 10

6 Reproducibility 11

1 Bound preservation with sign-changing sources (Direction B, PR5)

1.1 Setting and the AFC scheme

For $u_t - \nabla \cdot (\boldsymbol{K}\nabla u) = q$ with Dirichlet data g, the high-order FV operator A_H (reconstructed or non-orthogonally corrected flux) is consistent but not an M-matrix. Thermalio's AFC solver (PRs 2–4) uses the low-order operator $\mathsf{A}_L = \mathsf{A}_H + \mathsf{D}$, with $d_{ij} = -\max(0, a_{ij}, a_{ji})$ off the diagonal and zero row sums, which is an M-matrix with zero row sums. The difference is written as antisymmetric edge fluxes $f_{ij} = d_{ij}(u_i - u_j)$. One backward-Euler step solves

$$(\mathsf{D} + \Delta t\, \mathsf{A}_L)u^{n+1} = \mathsf{D}(u^n + \Delta t\, q^{n+1}) + \Delta t \sum_j \alpha_{ij} f_{ij}(u^{n+1}), \qquad \text{identity rows on Dirichlet cells,} \tag{1.1}$$

with Zalesak limiter coefficients $\alpha_{ij} \in [0,1]$ chosen so that the anti-diffusive correction keeps each node inside bounds $[u_i^{\min}, u_i^{\max}]$ ($\alpha \equiv 0$ gives the low-order solution u^L; $\alpha \equiv 1$ gives the high-order one). The scheme's properties are therefore determined by the bounds.

Why the PR4 global limiter fails with sources. PR4 sets $u^{\min} \equiv m$, $u^{\max} \equiv M$, where m and M are the extrema of the initial and boundary data. This is rigorous only if the continuous problem obeys $m \le u \le M$, i.e. for $q \equiv 0$ or sign-definite q. With a sign-changing source the true solution leaves $[m, M]$. Wherever the low-order solution exceeds M, the Zalesak capacity $Q_i^+ \propto \max(M - u_i, 0)$ is zero, so the correction is switched off there and the scheme falls back to the low-order (inconsistent on skewed meshes) operator.

1.2 The source envelope

Proposition 1.1 (Discrete comparison envelope)**.** *Let A_L be an M-matrix with zero row sums and $\mathsf{D} > 0$ diagonal. Define $M^0 = \max(\max_i u_i^0, \max g^0)$, $m^0 = \min(\min_i u_i^0, \min g^0)$ and*

$$\begin{aligned} M^{n+1} &= \max\Big(M^n + \Delta t \max_{i\in\mathcal{I}}(q_i^{n+1})^+,\ \max g^{n+1}\Big), \\ m^{n+1} &= \min\Big(m^n - \Delta t \max_{i\in\mathcal{I}}(q_i^{n+1})^-,\ \min g^{n+1}\Big). \end{aligned} \tag{1.2}$$

If $m^n \le u^n \le M^n$, then the low-order solution satisfies $m^{n+1} \le u^L \le M^{n+1}$.

Proof. Let $v \equiv M^{n+1}$ (a constant vector). Because the row sums are zero, $(\mathsf{D} + \Delta t \mathsf{A}_L)v = \mathsf{D}v$. On interior rows, $|\Omega_i| M^{n+1} \ge |\Omega_i|(M^n + \Delta t\, q_i^{n+1}) \ge |\Omega_i|(u_i^n + \Delta t\, q_i^{n+1})$. On Dirichlet rows, $v_i = M^{n+1} \ge g_i$. So v is a discrete supersolution, and the system matrix with identity Dirichlet rows is inverse-nonnegative (M-matrix), which gives $u^L \le v$. The lower bound is symmetric. □

The continuous analogue holds too: $w(t) = M^0 + \int_0^t \max_x q^+ \,\mathrm{d}s$ satisfies $w_t \geq q$ and $w \geq g$, so $u \leq w$ by the parabolic comparison principle. The `source_envelope` limiter enforces $u^{\min} \equiv m^{n+1}$, $u^{\max} \equiv M^{n+1}$ in (1.1).

- It never clips the true solution, because the true solution satisfies the same type of bound.
- It does not clip smooth interior extrema, which lie inside the envelope. This is the mechanism that gave PR4 its accuracy.
- For $q \equiv 0$ it reduces to PR4 exactly (verified to 10^{-13} in the unit tests).

1.3 Source-local bounds

Lemma 1.2 (Local convex-combination property of the low-order step)**.** *On an interior row with $\ell_{ij} = -(\mathsf{A}_L)_{ij} \geq 0$, the low-order solution satisfies*

$$u_i^L = \frac{|\Omega_i|\,\tilde{u}_i + \Delta t \sum_j \ell_{ij} u_j^L}{|\Omega_i| + \Delta t \sum_j \ell_{ij}}, \qquad \tilde{u}_i = u_i^n + \Delta t\, q_i^{n+1},$$

a convex combination of the source-advanced old value $\tilde{u}_i$ and the new neighbour values.

Source-driven extrema are therefore already present in u^L. The `source_local` limiter takes $[u_i^{\min}, u_i^{\max}]$ as the range of u^L over the stencil of i. The high-order correction may then sharpen features but not create extrema beyond those the monotone, source-including scheme produces. Since u^L lies inside the envelope (proposition 1.1), these bounds are at least as strict. As with all local limiters, smooth extrema are clipped unless the smoothness relaxation of PR3 is enabled; the relaxed variant is not strictly bound-preserving.

1.4 Results

Table 1: Relative L^2 error for the growing manufactured solution (fig. 1a).

scheme	$n = 12$	24	48	96
high-order (unlimited)	5.48e−3	2.09e−3	7.72e−4	2.38e−4
low-order M-matrix	6.75e−3	8.22e−3	8.28e−3	7.87e−3
PR4 global (auto bounds)	5.56e−3	4.38e−3	4.51e−3	4.60e−3
source envelope	5.48e−3	2.09e−3	7.72e−4	2.38e−4
source local, strict	6.93e−3	4.12e−3	2.74e−3	4.21e−3
source local, relaxed	5.48e−3	2.09e−3	7.72e−4	1.19e−3

On the steep problem (fig. 1b–d) the high-order scheme exceeds the envelope by up to 0.063, and so does the relaxed local limiter (0.061). The envelope, strict local, PR4 and low-order schemes stay inside it. The sink-driven minimum (≈ -0.028 for the envelope limiter) is *not* clipped. A fine-mesh high-order reference leaves the envelope by only 4.6×10^{-4}: the violation is a discretization artifact that vanishes under refinement. The L^2 errors against that reference (0.14–0.18) are dominated by the coarse resolution of the discontinuous data and do not separate the schemes; accuracy is assessed by table 1.

Assessment. The envelope limiter is the recommended default for sign-changing sources. It is rigorous, as accurate as the unlimited scheme, and inexpensive. Its bounds are global and can be loose, so it does not suppress small *local* oscillations that stay within the envelope. The strict local limiter does, at first-order accuracy.

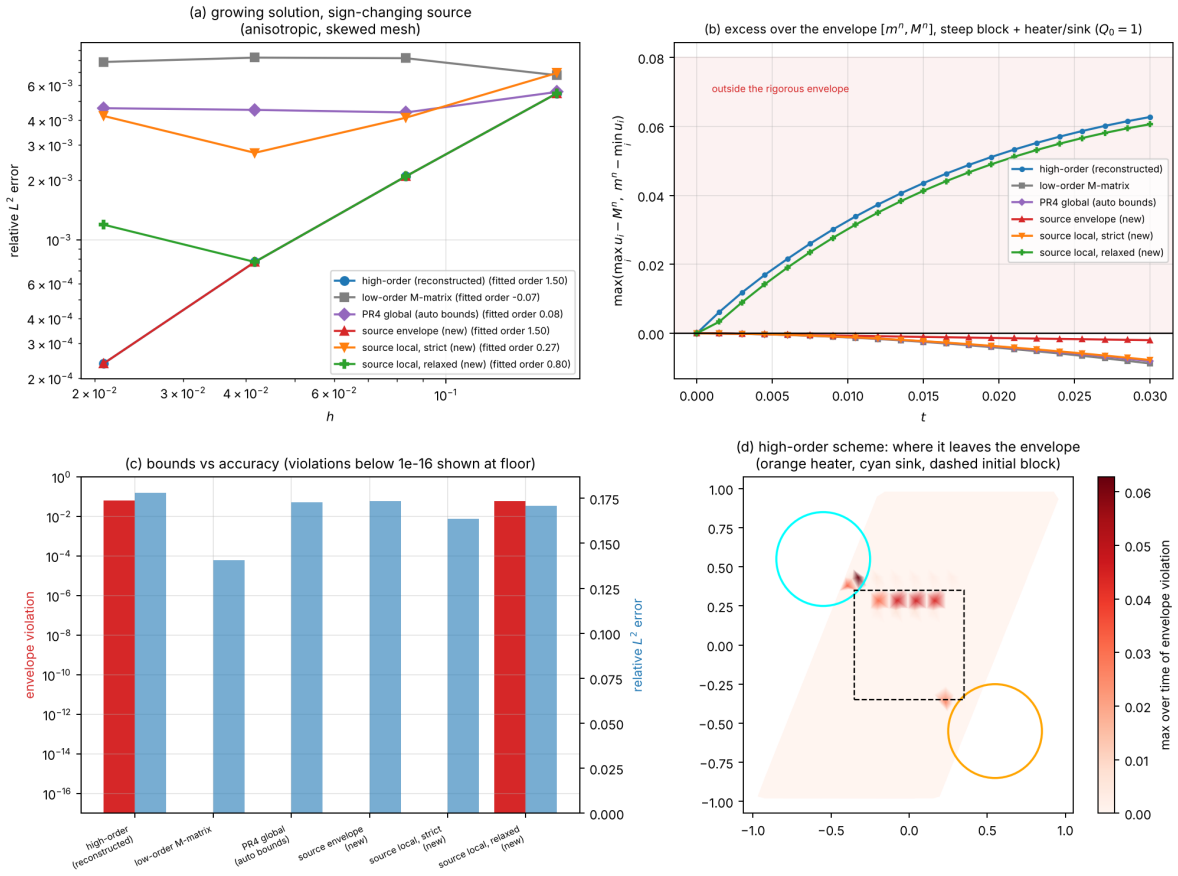


Figure 1: (a) Manufactured solution $u = (0.2 + 4t)\sin \pi x \sin \pi y$ under a sign-changing source. It grows from $[-0.2, 0.2]$ to $[-0.4, 0.4]$, outside the data range. Anisotropic tensor (ratio 5, 30°), skewed mesh. (b) Excess over the rigorous envelope for a steep block with a heater and a sink (ratio 20). (c) Maximum violation and error against a fine-mesh reference. (d) Where the high-order scheme leaves the envelope: at block corners aligned with the anisotropy.

2 Consistent Dirichlet boundary fluxes (Direction C, capability 7)

2.1 Reaction flux

Thermalio imposes Dirichlet data by replacing the equations of the boundary-cell band $\mathcal{B}$ with $u_i = g_i$, so the heat crossing $\partial\Omega$ is never computed. Following the consistent-flux idea of Hughes et al. [4], we evaluate the *unconstrained* cell balance on $\mathcal{B}$ at the computed solution:

$$r_i^{n+1} = |\Omega_i|\,\mathcal{T}_i^{n+1} + (\mathsf{K}u^{n+1})_i - |\Omega_i| q_i^{n+1} - b_i^{n+1}, \qquad i \in \mathcal{B}, \tag{2.1}$$

where $\mathcal{T}_i$ is the scheme's own discrete time term (e.g. $(H_i^{n+1} - H_i^n)/k$ for the enthalpy solver; $\tau D^2 u + c_{\text{app}} D^1 u$ for the apparent-capacity Cattaneo–Stefan solver). r_i is the heat flow into cell i through $\partial\Omega$ needed to hold $u_i = g_i$.

Proposition 2.1 (Exact global balance)**.** *If the FV operator is conservative (each face flux is added to one cell and subtracted from its neighbour, so that* $\mathbf{1}^\top \mathsf{A} = 0$*), then summing the cell equations gives*

$$\sum_i |\Omega_i| \mathcal{T}_i^{n+1} = \sum_i |\Omega_i| q_i^{n+1} + \sum_i b_i^{n+1} + \sum_{i \in \mathcal{B}} r_i^{n+1} + \sum_{i \in \mathcal{I}} \rho_i^{n+1}, \tag{2.2}$$

where ρ_i is the residual of interior equations (zero for converged nonlinear solves).

Thermalio's TPFA, non-orthogonal and reconstructed operators have $\mathbf{1}^\top \mathsf{A} = 0$ to round-off; the audit records this.

2.2 Audit identity for the apparent-capacity Cattaneo–Stefan scheme

With $D^2u = (u^{n+1} - 2u^n + u^{n-1})/\Delta t^2$ and $D^1u = (u^{n+1} - u^{n-1})/(2\Delta t)$, multiplying by Δt and summing over steps, the relaxation term telescopes:

$$\underbrace{\tau \sum_i |\Omega_i| \frac{(u^N - u^{N-1}) - (u^0 - u^{-1})}{\Delta t}}_{\text{relaxation storage } R} + \underbrace{\Delta t \sum_n \sum_i |\Omega_i| c_{\text{app}}(u^{n+1}) D^1 u}_{\text{capacity work } C} = Q + F + \varrho, \tag{2.3}$$

with source energy Q, reconstructed boundary energy $F = \Delta t \sum_n \sum_{\mathcal{B}} r^{n+1}$ and unconverged-step residual ϱ. The enthalpy audit compares the true enthalpy change $\Delta E = \sum_i |\Omega_i|[H(u_i^N) - H(u_i^0)]$ with $Q + F - R$:

$$\Delta E - (Q + F - R) = \underbrace{(\Delta E - C)}_{\text{capacity defect}} + \varrho. \tag{2.4}$$

Equation (2.3) holds to 10^{-15} in every run (unit test and study).

Lemma 2.2 (Latent-heat jump of the Dirichlet band)**.** *If a band cell is initialized at T_0 (liquid) while g lies below the solidus, the cell jumps to g in the first step. Its true enthalpy change is $c(g - T_0) - L$, while the apparent-capacity time term (with $c_{\text{app}}(g) = c$) accounts for only $c(g - T_0)$. The band's contribution to the capacity defect is therefore $-L \sum_{i \in \mathcal{B}_{\text{jump}}} |\Omega_i|$, independently of Δt.*

We therefore also report an *interior* audit, using the band-to-interior flux $-\sum_{i \in \mathcal{I}} (\mathsf{A}u)_i$, which isolates the scheme's own defect.

2.3 Results

Table 2: Audit decomposition, apparent-capacity Cattaneo–Stefan solver, 24-cell mesh, 100 steps (energy units). Lemma 2.2 predicts band jumps $-L|\mathcal{B}_{\text{jump}}| = -0.1667$ and -0.612.

	cryosurgery	quench
enthalpy change ΔE	-0.639	-1.864
reconstructed boundary energy F	-0.475	-1.276
relaxation storage R	-0.0006	-0.0023
band latent jump	-0.1667	-0.612
interior capacity defect	$+0.085$	$+0.065$
unconverged-step residual ϱ	-0.084	-0.043
closure residual	-0.165 (26%)	-0.590 (32%)
identity (2.3) residual	$< 10^{-15}$	$< 10^{-15}$
enthalpy solver closure	-1.1×10^{-16}	2.2×10^{-16}

As Δt decreases from 0.02 to 0.00125, the interior capacity defect for cryosurgery falls from 0.18 to 0.017 and the unconverged residual from 0.23 to 0.008, while the band jump stays at -0.1667. The production application runners (`direction_c_applications.py`) now use this audit. At their default 40×40 and 36×18 resolutions they report relative closure residuals of -2.1% (cryosurgery) and -12.5% (quench), where the previous inferred audit reported exactly zero.

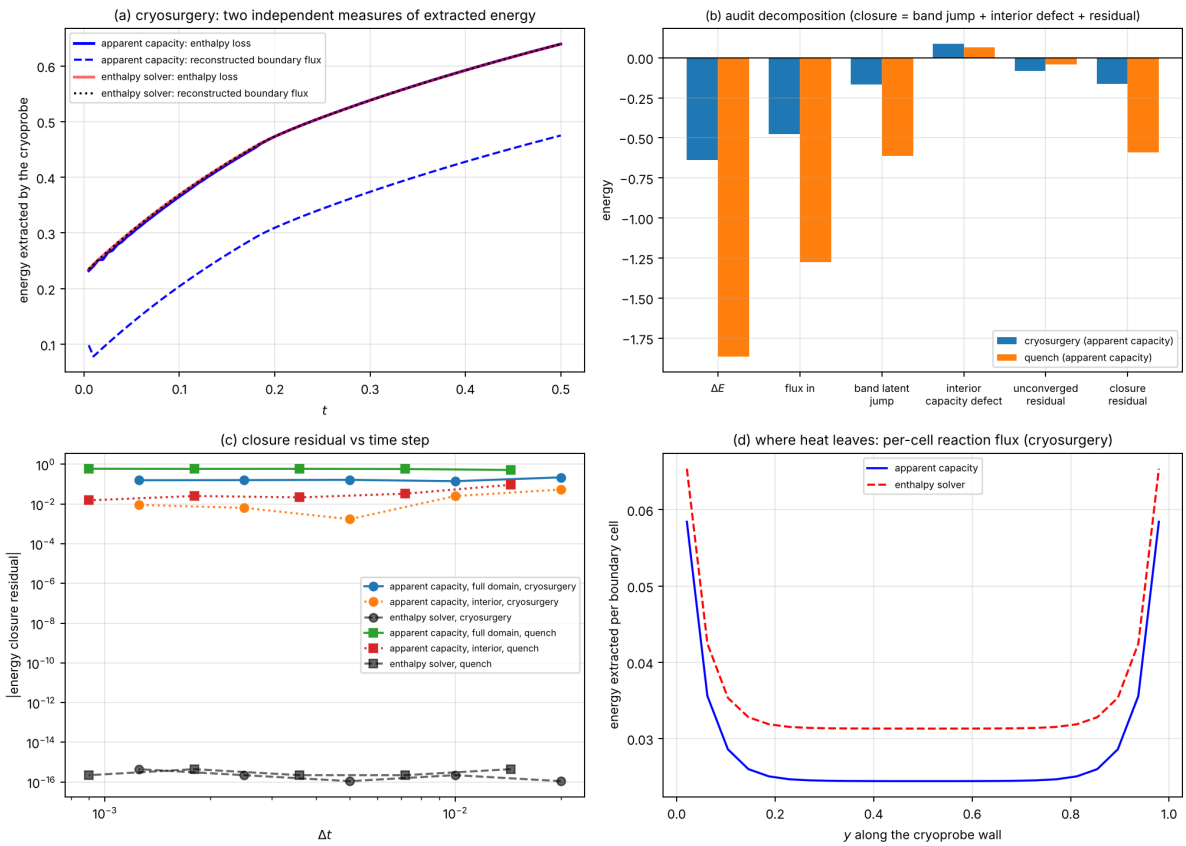


Figure 2: (a) Cryosurgery: energy extracted measured two independent ways. For the enthalpy solver the reconstructed flux and the enthalpy loss coincide; for the apparent-capacity solver they differ by 26%. (b) Exact decomposition of the closure residual. (c) Closure residual vs. Δt: full domain (band jump, Δt-independent), interior only (decreasing), enthalpy solver (10^{-16}). (d) Per-cell reaction flux along the cryoprobe.

3 Cross-code validation against external codes (Direction A)

3.1 Codes and protocol

- **Thermalio**: cell-centred FV (reconstructed flux) on square and skewed (0.3) quadrilateral meshes. Dirichlet data are imposed on the boundary-cell band at cell centres; Robin data use a one-sided face closure.
- **FiPy 4.0.3** [5]: NIST cell-centred FV on `Grid2D`. Dirichlet values are imposed on faces; a rank-2 face coefficient is used for the tensor; SciPy LU backend.
- **scikit-fem 12.0.2** [6]: continuous Galerkin P1/P2 on a structured triangulation. Dirichlet data are imposed nodally, Robin data weakly.

All codes share no source with Thermalio's solvers (`heat_solver/external_codes.py` only calls their public APIs) and use backward Euler with the same $\Delta t = 2.5 \times 10^{-4}$, $t_f = 0.05$. The temporal error ($\approx 5 \times 10^{-6}$) is therefore common and small. FV errors are discrete relative L^2 norms at cell centres; FEM errors are exact L^2 norms by quadrature. Cross-code differences are evaluated at Thermalio's cell centres (FiPy's centres coincide; FEM solutions are interpolated with `basis.probes`).

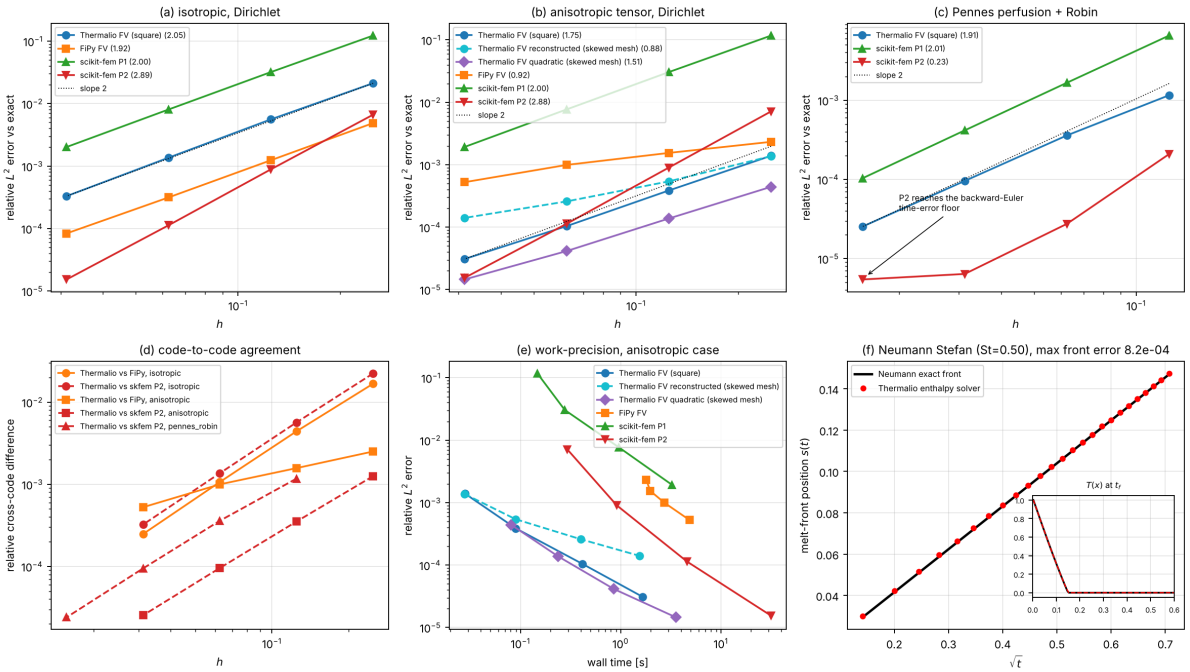


Figure 3: (a)–(c) Convergence to the exact solution; legend values are finest-pair orders. (d) Code-to-code differences. (e) Work–precision for the anisotropic case. (f) Enthalpy Stefan solver vs. Neumann's similarity solution.

Table 3: Relative L^2 error at $n = 64$ (finest-pair observed order in brackets).

code	isotropic	anisotropic (ratio 5)	Pennes + Robin
Thermalio, square	3.30e−4 (2.05)	3.09e−5 (1.75)	2.54e−5 (1.91)
Thermalio, skewed	–	1.41e−4 (0.88)	–
FiPy	8.34e−5 (1.92)	5.25e−4 (0.92)	–
scikit-fem P1	2.02e−3 (2.00)	1.93e−3 (2.00)	1.03e−4 (2.01)
scikit-fem P2	1.53e−5 (2.89)	1.54e−5 (2.88)	5.45e−6 (time floor)

3.2 Findings

1. **Agreement.** All codes converge at their nominal orders on the isotropic and Robin/Pennes problems. The Thermalio–FiPy and Thermalio–scikit-fem (P2) differences decay at second order to 2.5×10^{-4} and 2.5×10^{-5}–3.2×10^{-4}. This is independent-code verification of Thermalio's diffusion, Robin and perfusion discretizations.

2. **Error constants.** On the isotropic problem Thermalio's error is 4× FiPy's at equal cell count. The difference comes from the Dirichlet convention: FiPy imposes g on boundary faces, while Thermalio holds the whole boundary-cell band at $g(\boldsymbol{x}_i)$, which is still second-order consistent but shifts the effective boundary by half a cell.

3. **Anisotropy.** With a rotated tensor, Thermalio's reconstructed flux stays close to second order (orders $1.85, 1.90, 1.75$) and is 17× more accurate than FiPy at $n = 64$. FiPy's rank-2 face coefficient on `Grid2D` converges at order 0.6–0.9 in this test. We report this as observed behaviour of the documented API, not as a general statement about FiPy.

4. **Limitation of Thermalio (now fixed).** On the skewed mesh the same anisotropic problem converged at orders $1.34 \to 1.07 \to 0.88$ with the `reconstructed` flux. Section 4 diagnoses the cause and introduces the quadratic flux. In this cross-code protocol it reaches 1.45×10^{-5}

at $n = 64$ ("Thermalio FV quadratic (skewed mesh)" in fig. 3), $10\times$ below the `reconstructed` flux (1.41×10^{-4}). Its finest-pair order here (1.51) is capped by the shared backward-Euler time floor; with Crank–Nicolson it is 2.00 (table 4).

5. **Time-error floor.** P2 reaches the common backward-Euler floor of $\approx 5 \times 10^{-6}$ on the Robin problem, which is why its last observed order is 0.23.

6. **Cost.** At $n = 64$ the wall times are: Thermalio 1.6 s, scikit-fem P1 3.5 s, FiPy 5–6 s, scikit-fem P2 34 s. P2 is the most accurate per degree of freedom on these smooth problems.

3.3 Neumann's Stefan problem

A solid at the melting temperature $T_m = 0$ is melted by a wall at $T_w = 1$. With Stefan number $\mathrm{St} = c(T_w - T_m)/L = 0.5$, the front is $s(t) = 2\lambda\sqrt{\alpha t}$, where $\lambda e^{\lambda^2} \operatorname{erf}\lambda = \mathrm{St}/\sqrt{\pi}$ ($\lambda = 0.4648$), and the liquid temperature is $T = T_w[1 - \operatorname{erf}(x/2\sqrt{\alpha t})/\operatorname{erf}\lambda]$ [7]. The enthalpy solver ($\Delta x = 1/160$, nearly sharp ramp of width 10^{-3}) tracks the front with a maximum error of $8.2 \times 10^{-4} = 0.13\,\Delta x$ over $t \in [0.02, 0.5]$ (fig. 3f).

4 Restoring second order on non-orthogonal meshes: the quadratic flux

4.1 Diagnosis

Three findings, each separated by its own experiment:

1. *Part of the reported loss was the time step.* With backward Euler ($\Delta t = 2.5 \times 10^{-4}$) the finest anisotropic errors ($\sim 10^{-5}$) reach the temporal floor ($\approx 6 \times 10^{-6}$). All results below use Crank–Nicolson with $\Delta t = 10^{-4}$ and $t_f = 0.02$, which removes it.

2. *The `reconstructed` flux is genuinely defective.* Its face gradient is the plain average $\frac{1}{2}(\nabla_h u_i + \nabla_h u_j)$ of least-squares (LSQ) cell gradients. It has no compact two-point term, so the stencil is wide and weakly coupled, and its deep-interior error stops converging on the sheared mesh. The TPFA flux with non-orthogonal correction (a two-point term along $\boldsymbol{d}$ plus a gradient correction) is second order on the sheared mesh.

3. *All linearly exact gradient schemes are inconsistent on rough meshes.* An LSQ gradient that is exact only for linear fields has an $O(h)$ error on non-symmetric stencils. The face flux is then only $O(h)$ accurate, and the discrete operator has an $O(1)$ truncation error: applied to the cell averages of a quadratic it misses the source by 10^{-3} to 10^{-1} (fig. 4, top right).

4.2 Formulation

2-exact reconstruction. On each cell we seek a quadratic

$$p_i(\boldsymbol{x}) = c_i + \boldsymbol{g}_i \cdot (\boldsymbol{x} - \boldsymbol{x}_i) + \tfrac{1}{2}(\boldsymbol{x} - \boldsymbol{x}_i)^\top \mathsf{H}_i (\boldsymbol{x} - \boldsymbol{x}_i) \tag{4.1}$$

whose cell average over Ω_i equals $\bar{u}_i$. With centroid $\boldsymbol{x}_i$ and central second moments $\mathsf{M}_i = |\Omega_i|^{-1} \int_{\Omega_i} (\boldsymbol{x} - \boldsymbol{x}_i)(\boldsymbol{x} - \boldsymbol{x}_i)^\top$ (closed-form polygon formulas), the average of p_i over a neighbour cell j gives the fitting equations

$$\bar{u}_j - \bar{u}_i = \boldsymbol{g}_i \cdot \boldsymbol{\Delta}_{ij} + \tfrac{1}{2}\boldsymbol{\Delta}_{ij}^\top \mathsf{H}_i \boldsymbol{\Delta}_{ij} + \tfrac{1}{2}\,\mathsf{H}_i : (\mathsf{M}_j - \mathsf{M}_i), \qquad \boldsymbol{\Delta}_{ij} = \boldsymbol{x}_j - \boldsymbol{x}_i, \tag{4.2}$$

for all j in the vertex-ring stencil (extended by a second ring for boundary cells with fewer than 7 neighbours). They are solved by weighted least squares (weights $|\boldsymbol{\Delta}_{ij}|^{-2}$, columns scaled by h and h^2) for $(\boldsymbol{g}_i, \mathsf{H}_i)$, which are linear in $\bar{u}$. The moment term is what makes the fit exact for quadratics given *cell averages*; omitting it leaves an $O(h)$ gradient error.

Face flux. At the midpoint $\boldsymbol{x}_f$ of face $f = (i, j)$ with unit normal $\boldsymbol{n}$ (from i to j) and $\boldsymbol{d} = \boldsymbol{x}_j - \boldsymbol{x}_i$,

$$\boldsymbol{G}_f = \tfrac{1}{2}\big[\boldsymbol{g}_i + \mathsf{H}_i(\boldsymbol{x}_f - \boldsymbol{x}_i) + \boldsymbol{g}_j + \mathsf{H}_j(\boldsymbol{x}_f - \boldsymbol{x}_j)\big], \tag{4.3}$$

$$J_f = \tfrac{1}{2}\big[(p_i\text{-predicted } \bar{u}_j - \bar{u}_i) + (p_j\text{-predicted } \bar{u}_j - \bar{u}_i)\big], \tag{4.4}$$

$$F_f = -|f|\,\boldsymbol{n}^\top \mathsf{K}\boldsymbol{G}_f - |f|\,T_f\big[(\bar{u}_j - \bar{u}_i) - J_f\big], \qquad T_f = \frac{\boldsymbol{n}^\top \mathsf{K}\boldsymbol{n}}{\boldsymbol{n}\cdot\boldsymbol{d}} > 0. \tag{4.5}$$

F_f is the outward flux of $-\mathsf{K}\nabla u$ from i; it is added to row i and subtracted from row j, so the scheme is conservative. The penalty restores compact two-point coupling (it is the TPFA term) but is measured against the jump the reconstruction *predicts*, so it vanishes whenever the data are reconstructed exactly.

Proposition 4.1 (Quadratic exactness)**.** *Let* K *be constant, faces straight and* u *a quadratic polynomial with exact cell averages* $\bar{u}$*. If every stencil has full rank for* (4.2)*, then* $F_f = -\int_f \boldsymbol{n}^\top \mathsf{K}\nabla u$ *exactly for every interior face. Hence* $\sum_{f\subset\partial\Omega_i} F_f = -\int_{\Omega_i} \nabla\cdot(\mathsf{K}\nabla u)$*, i.e. the discrete operator has zero truncation error on quadratics.*

Proof. For quadratic u, (4.2) holds exactly with $(\boldsymbol{g}_i, \mathsf{H}_i) = (\nabla u(\boldsymbol{x}_i), \nabla^2 u)$, so the least-squares solution is exact. Then $\boldsymbol{G}_f = \nabla u(\boldsymbol{x}_f)$, $J_f = \bar{u}_j - \bar{u}_i$, and the penalty vanishes. Since ∇u is linear along the straight face, the midpoint rule gives $|f|\,\boldsymbol{n}^\top \mathsf{K}\nabla u(\boldsymbol{x}_f) = \int_f \boldsymbol{n}^\top \mathsf{K}\nabla u$. □

Exactness for quadratics makes the flux $O(h^2)$ consistent for smooth u. For a stable conservative scheme this gives second-order convergence of the cell averages.

Consistent data. Because (4.2) effectively differentiates the data twice, an $O(h^2)$ inconsistency δ in the cell values (e.g. Dirichlet band values sampled at centroids instead of averaged) produces Hessian errors $O(\delta/h^2) = O(1)$ near the boundary. Measured on perturbed quadrilaterals, this reduces the observed order from ≈ 2 to 0.38. The solver therefore integrates the source and Dirichlet data over each cell with a degree-4 triangle-fan quadrature when the quadratic flux is selected (`cell_average_data`, default on for this flux only). It also provides `cell_average()` for initial data and error measurement. The two-point schemes are insensitive to this and keep their previous behaviour, so no regression baseline changes.

4.3 Results

Table 4: Relative L^2 error on the finest mesh (finest-pair observed order). Quadratic truncation error in the last column block.

mesh	reconstructed	TPFA + corr.	**quadratic**	truncation (quad. / TPFA)
skewed ($n = 128$)	3.39e−5 (0.85)	1.03e−6 (1.98)	1.00e−6 (2.00)	2e−14 / 1e−13
hexagonal ($n = 128$)	2.60e−5 (1.14)	1.63e−5 (1.01)	4.17e−6 (1.93)	5e−15 / 4e−3
perturbed ($n = 128$)	2.63e−4 (0.24)	1.43e−4 (0.94)	3.29e−6 (2.03)	1e−14 / 2e−2
tiled ($n = 96$)	6.40e−4 (0.22)	3.57e−4 (0.75)	2.77e−5 (1.92)	7e−15 / 3e−2

On the sheared mesh the corrected TPFA is already second order, because a uniform shear keeps the face midpoint at the midpoint of the centroids and makes its gradient correction exact for quadratics. The quadratic flux matches it there and is the only option that stays second order on hexagonal, randomly perturbed and mixed polygonal meshes, with finest-mesh errors 4–80× smaller. The AFC limiters accept it as their high-order target (`flux_discretization="quadratic"`) and remain strictly bounded (unit test).

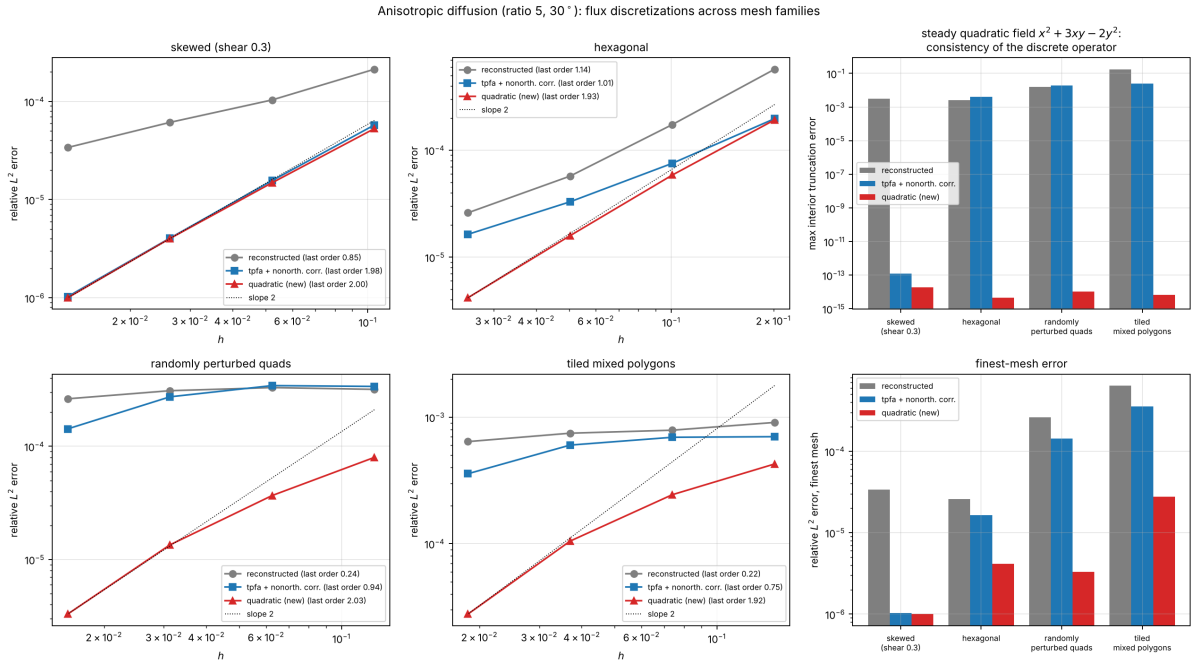


Figure 4: Anisotropic diffusion (ratio 5, 30°), consistent cell-average data, Crank–Nicolson. Four mesh families; top right: maximum interior truncation error of the steady operator on $x^2 + 3xy - 2y^2$; bottom right: finest-mesh error.

5 Limitations and next steps

- The envelope bounds accumulate $\Delta t \max q^{\pm}$ and can become loose for long runs; a sharper envelope would solve the source-only problem locally. The relaxed local limiter is not strictly bound-preserving.

- The reconstructed-flux audit attributes energy correctly but does not cure the apparent-capacity defects it reveals; the enthalpy solver does. The Dirichlet-band latent jump is a modelling artefact of initial data that are inconsistent with the boundary data; consistent initial data or face-based Dirichlet conditions would remove it.

- The quadratic flux is not provably coercive. Numerically, the symmetric part of the operator is positive definite and the time-stepping spectrum lies in the right half plane on all four mesh families for anisotropy up to 10^4 (unit tests). It is not monotone (use it as the high-order target of the AFC limiters for bounds) and has ≈ 21 non-zeros per row (vs. 5 for TPFA). Its geometry is precomputed once (6 s for 16 384 cells) and re-assembly is vectorized (0.12 s), which makes temperature-dependent (tensor) conductivity practical. This is supported, with face temperatures from the quadratic reconstruction, and converges at order ≈ 1.9 on skewed and hexagonal meshes. The same work found and fixed a first-order defect in the solver's nonlinear Crank–Nicolson step (the explicit half used $\mathsf{A}(u^{n+1})$).

- A measured gap analysis against FiPy, scikit-fem, FEniCS and deal.II (2D only, Python-loop setup cost, no preconditioned or parallel solvers, band Dirichlet, solver-specific APIs) is in `docs/research/gap_analysis_vs_established_codes.md`.

- The external comparison is for linear problems. A nonlinear (Stefan) cross-code comparison would require writing an enthalpy model in the external code, which would no longer be independent. The Neumann solution serves as the external reference for the nonlinear case.

6 Reproducibility

```
python -m pip install -r requirements-external.txt
export MPLCONFIGDIR=/tmp/matplotlib
python -m pytest tests/test_afc.py tests/test_boundary_flux.py \
       tests/test_external_codes.py tests/test_quadratic_flux.py -q
python quadratic_flux_study.py           # ~4 min
python dmp_source_study.py               # ~1 min
python direction_c_flux_audit_study.py   # ~1 min
python external_code_comparison.py       # ~3 min
```

CI runs the cross-code tests in a separate job that installs the pinned external codes from `requirements-external.txt`.

References

[1] D. Kuzmin, Explicit and implicit FEM-FCT algorithms with flux linearization, *J. Comput. Phys.* 228 (2009) 2517–2534.

[2] S. T. Zalesak, Fully multidimensional flux-corrected transport algorithms for fluids, *J. Comput. Phys.* 31 (1979) 335–362.

[3] X. Zhang and C.-W. Shu, On maximum-principle-satisfying high order schemes for scalar conservation laws, *J. Comput. Phys.* 229 (2010) 3091–3120.

[4] T. J. R. Hughes, G. Engel, L. Mazzei and M. G. Larson, The continuous Galerkin method is locally conservative, *J. Comput. Phys.* 163 (2000) 467–488.

[5] J. E. Guyer, D. Wheeler and J. A. Warren, FiPy: partial differential equations with Python, *Comput. Sci. Eng.* 11(3) (2009) 6–15.

[6] T. Gustafsson and G. D. McBain, scikit-fem: a Python package for finite element assembly, *J. Open Source Softw.* 5(52) (2020) 2369.

[7] H. S. Carslaw and J. C. Jaeger, *Conduction of Heat in Solids*, 2nd ed., Oxford University Press, 1959.